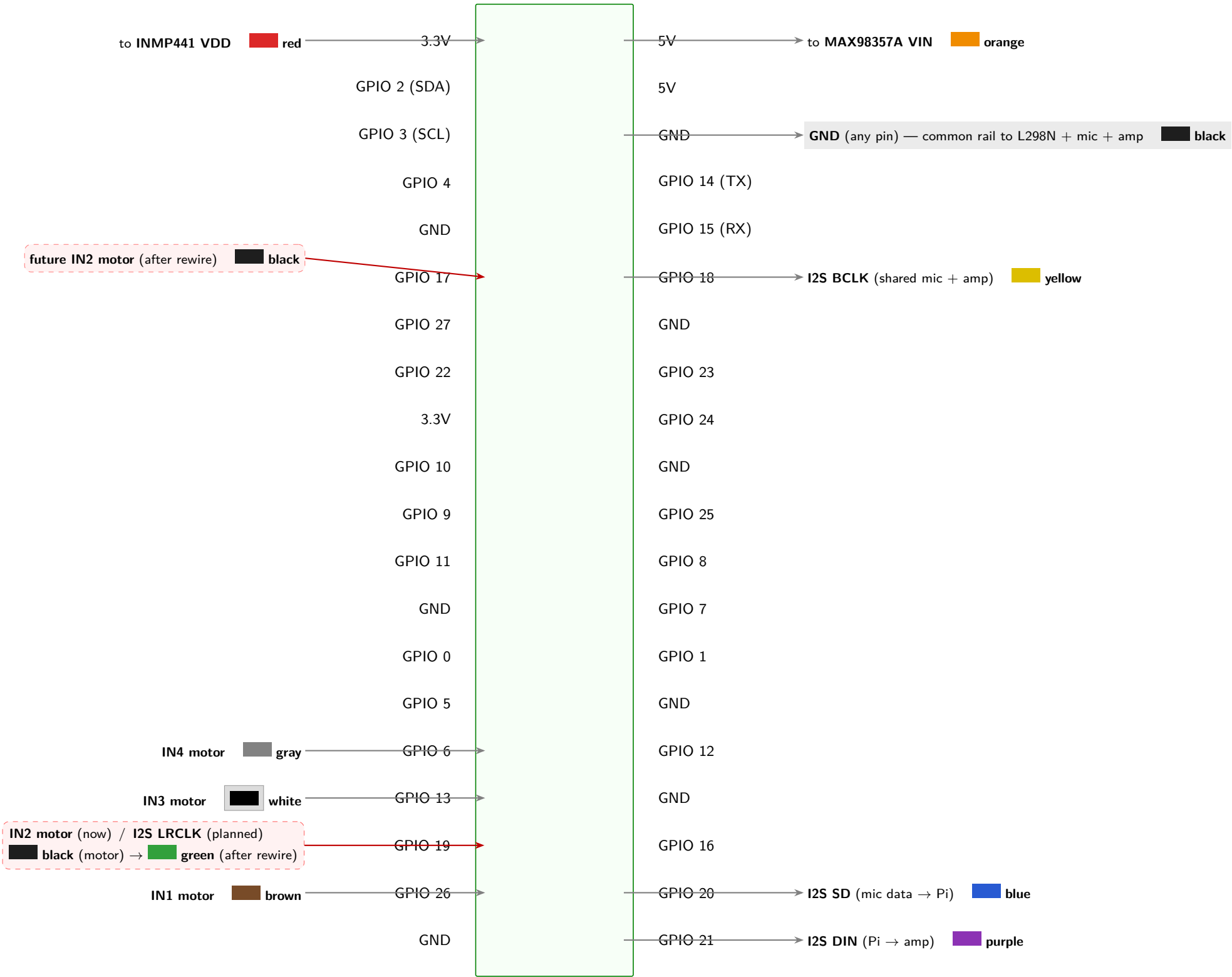


robothector — Raspberry Pi GPIO header & wire-colour reference

Raspberry Pi 40-pin GPIO header (top-down view)



Pin status (header colour)

- currently used (motor IN1–IN4)
- planned for Phase 4 audio
- conflict pin (must rewire IN2)
- power rail (3.3V or 5V)
- ground
- unused / available

Recommended Dupont wire colours

- red — 3.3V (mic VDD)
- orange — 5V (amp VIN)
- black — GND (any pin)
- yellow — I2S BCLK (shared)
- green — I2S LRCLK (shared)
- blue — I2S SD (mic data)
- purple — I2S DIN (amp data)
- brown — IN1 motor (Pi p37)
- black — IN2 motor (Pi p35)*
- white — IN3 motor (Pi p33)
- gray — IN4 motor (Pi p31)

*IN2 wire moves to Pi pin 11 (GPIO 17) when audio is added (frees GPIO 19 for LRCLK). NOT GPIO 16: claimed by I2S overlay for amp SD_MODE.

Pi-pin → destination map

Pi pin	Function	Goes to
1	3.3V	INMP441 VDD
2	5V	MAX98357A VIN
6, 9, 14, 20...	GND	L298N + mic + amp + spk
11	GPIO 17*	L298N IN2 (after rewire)
12	GPIO 18	mic SCK + amp BCLK
31	GPIO 6	L298N IN4
33	GPIO 13	L298N IN3
35	GPIO 19*	mic WS + amp LRC
37	GPIO 26	L298N IN1
38	GPIO 20	INMP441 SD
40	GPIO 21	MAX98357A DIN

* rewire: free GPIO 19 by moving IN2 to GPIO 17 (Pi pin 11) — GPIO 16 is unavailable (claimed by I2S overlay for amp SD_MODE).